

Security in Computing

Practice Questions — Complete Solutions

From-Scratch Topology Setup · Full CLI · Verification
Q1 – Q10 · University of Mumbai · Part 1 of 2

QUESTION 1

OSPF MD5 Authentication + NTP + Syslog

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
R1	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
R2	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
R3	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A (NTP)	NIC	192.168.1.5	255.255.255.0	192.168.1.1
PC-B (Syslog)	NIC	192.168.1.6	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1

■ TOPOLOGY SETUP (FROM SCRATCH)

Place devices: 3 routers (1941), Switch0 (2960), PCA, PCB, PCC. Cable: PCA→Fa0 of Switch0, PCB→Fa0 of Switch0, Switch0→R1 Gig0/0, R1 se0/1/0↔R2 se0/1/0 (serial DCE on R1), R2 se0/1/1↔R3 se0/1/0 (serial DCE on R2), R3 Gig0/0→Switch1, PCC→Switch1.

■ STEPS

Step 1 — Configure IPs on all devices

```
R1(config)#
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.1.1.1 255.255.255.252
clock rate 128000
no shutdown
exit
```

```
R2(config)#
interface se0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
interface se0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
```

```
R3(config)#
interface gig0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
exit
```

Step 2 — Configure OSPF on all routers

```
R1(config)#
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.3 area 0
exit
```

```
R2(config)#
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

```
R3(config)#
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

Step 3 — Verify OSPF is FULL, then add MD5 on all routers

Run 'show ip ospf neighbor' and wait until all adjacencies show FULL, then add authentication.

```
R1(config)#
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

```
R2(config)#
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
interface se0/1/1
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

```
R3(config)#
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

Step 4 — Enable NTP auth on PC-A (NTP Server)

Click PC-A → Services → NTP → On. Under Authentication click Enable, Key=1, Password=NTPpa55.

Step 5 — Configure NTP on all routers

```
R1(config)#
ntp server 192.168.1.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
ntp authentication-key 1 md5 NTPpa55
service timestamps log datetime msec
```

```
R2(config)#
ntp server 192.168.1.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
```

```
ntp authentication-key 1 md5 NTPpa55
service timestamps log datetime msec
```

```
R3(config)#
```

```
ntp server 192.168.1.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
ntp authentication-key 1 md5 NTPpa55
service timestamps log datetime msec
```

Step 6 — Configure Syslog on all routers (PC-B is Syslog server)

Click PC-B → Services → SYSLOG → On, then configure each router:

```
R1(config)#
```

```
logging host 192.168.1.6
```

```
R2(config)#
```

```
logging host 192.168.1.6
```

```
R3(config)#
```

```
logging host 192.168.1.6
```

■ VERIFICATION

```
R1# show ip ospf interface se0/1/0 → Message digest authentication enabled Key id 1
```

```
R1# show ip ospf neighbor → All neighbours in FULL state
```

```
R1# show ntp status → synchronized, stratum 2, reference 192.168.1.5
```

```
Enter/exit config on R1, check PC-B SYSLOG tab → %SYS-5-CONFIG_I entries appear
```

■ KEY POINTS

- Clock rate only on DCE end. Verify with: show controllers se0/1/0 — look for 'DCE'.
- MD5 key MUST be on both the OSPF process (area auth) AND the interface. Missing one = EXSTART.
- If ping to syslog server fails, syslog will never work. Fix reachability first.

QUESTION 2

Extended ACLs — PC1 FTP only, PC2 HTTP only

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	172.22.34.65	255.255.255.224	N/A
R1	gig0/1	172.22.34.97	255.255.255.240	N/A
R1	gig0/2	172.22.34.1	255.255.255.192	N/A
Server	NIC	172.22.34.62	255.255.255.192	172.22.34.1
PC1	NIC	172.22.34.66	255.255.255.224	172.22.34.65
PC2	NIC	172.22.34.98	255.255.255.240	172.22.34.97

■ TOPOLOGY SETUP

Place: 1 router (1941), Switch0 (for PC1), Switch1 (for Server), Switch2 (for PC2). Cable: PC1→Switch0→R1 Gig0/0, Server→Switch1→R1 Gig0/2, PC2→Switch2→R1 Gig0/1.

■ STEPS

Step 1 — IP addressing on R1

Wildcard /27=0.0.0.31, /28=0.0.0.15, /26=0.0.0.63. Enable HTTP and FTP on Server (Services → HTTP on, FTP on with user cisco/cisco).

```
R1(config)#
interface gig0/0
ip address 172.22.34.65 255.255.255.224
no shutdown
exit
interface gig0/1
ip address 172.22.34.97 255.255.255.240
no shutdown
exit
interface gig0/2
ip address 172.22.34.1 255.255.255.192
no shutdown
exit
```

Step 2 — ACL 100 — PC1 FTP + ICMP only (applied on Gig0/0 inbound)

```
R1(config)#
access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ftp
access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62
interface gig0/0
ip access-group 100 in
exit
```

Step 3 — Named ACL HTTP_ONLY — PC2 HTTP + ICMP only (applied on Gig0/1 inbound)

```
R1(config)#
ip access-list extended HTTP_ONLY
permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www
permit icmp 172.22.34.96 0.0.0.15 host 172.22.34.62
exit
interface gig0/1
ip access-group HTTP_ONLY in
exit
```

■ VERIFICATION

PC1 ping Server → Success (ICMP permitted)

```
PC1 ftp 172.22.34.62 → Connected (FTP permitted)
PC1 browser http://172.22.34.62 → Timeout (HTTP blocked)
PC2 browser http://172.22.34.62 → Page loads (HTTP permitted)
PC2 ftp 172.22.34.62 → Timeout (FTP blocked)
R1# show access-lists → permit lines show match counts > 0
```

■ KEY POINTS

- Named ACL is case-sensitive: HTTP_ONLY must be typed exactly.
- eq ftp = port 21. eq www = port 80. ICMP has no port number.
- Extended ACLs placed closest to SOURCE.

QUESTION 3

IPv6 Static Routing + Block ICMP on R3

■ ADDRESSING TABLE

Device	Interface	IPv6 Address/Prefix	Default GW
PC1	NIC	2001:DB8:1:10::10/64	FE80::1
PC2	NIC	2001:DB8:1:11::11/64	FE80::1
R1	gig0/0	2001:DB8:1:10::1/64	—
R1	gig0/1	2001:DB8:1:11::1/64	—
R1	se0/1/0	2001:DB8:1:1::1/64	—
R2	se0/1/0	2001:DB8:1:1::2/64	—
R2	se0/1/1	2001:DB8:1:2::2/64	—
R3	gig0/0	2001:DB8:1:30::1/64	—
R3	se0/1/0	2001:DB8:1:2::1/64	—
Server	NIC	2001:DB8:1:30::30/64	FE80::3

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch1 (PC1, PC2 side), Switch2 (Server side). Use serial cables for R1-R2 and R2-R3 links.

■ STEPS

Step 1 — Enable IPv6 routing and configure all interfaces

```
R1(config)#
```

```
ipv6 unicast-routing
interface gig0/0
ipv6 address 2001:DB8:1:10::1/64
ipv6 address FE80::1 link-local
no shutdown
exit
interface gig0/1
ipv6 address 2001:DB8:1:11::1/64
ipv6 address FE80::1 link-local
no shutdown
exit
interface se0/1/0
ipv6 address 2001:DB8:1:1::1/64
clock rate 128000
no shutdown
exit
```

```
R2(config)#
```

```
ipv6 unicast-routing
interface se0/1/0
ipv6 address 2001:DB8:1:1::2/64
no shutdown
exit
interface se0/1/1
ipv6 address 2001:DB8:1:2::2/64
clock rate 128000
no shutdown
exit
```

```
R3(config)#
```

```
ipv6 unicast-routing
interface gig0/0
ipv6 address 2001:DB8:1:30::1/64
```

```
ipv6 address FE80::3 link-local
no shutdown
exit
interface se0/1/0
ipv6 address 2001:DB8:1:2::1/64
no shutdown
exit
```

Step 2 — Configure static routes on each router

```
R1(config)#
ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:1::2
ipv6 route 2001:DB8:1:2::/64 2001:DB8:1:1::2
```

```
R2(config)#
ipv6 route 2001:DB8:1:10::/64 2001:DB8:1:1::1
ipv6 route 2001:DB8:1:11::/64 2001:DB8:1:1::1
ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:2::1
```

```
R3(config)#
ipv6 route 2001:DB8:1:10::/64 2001:DB8:1:2::2
ipv6 route 2001:DB8:1:11::/64 2001:DB8:1:2::2
```

Step 3 — IPv6 ACL on R3 to block ICMP to Server (outbound on Gig0/0)

```
R3(config)#
ipv6 access-list BLOCK_ICMP
deny icmp any any
permit ipv6 any any
exit
interface gig0/0
ipv6 traffic-filter BLOCK_ICMP out
exit
```

■ VERIFICATION

```
PC1 ping 2001:DB8:1:30::30 → Timeout (ICMP blocked by ACL)
PC1 browser http://[2001:DB8:1:30::30] → Page loads (HTTP not blocked)
R3# show ipv6 interface gig0/0 → Outbound access list is BLOCK_ICMP
```

■ KEY POINTS

- "ipv6 unicast-routing" required on ALL routers — without it IPv6 packets are not forwarded.
- "ipv6 traffic-filter" not "ip access-group" for IPv6 ACLs.
- Static route next-hop must be reachable — ping the next-hop first.

QUESTION 4

OSPF MD5 + Local AAA on R1

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
PC0	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.3	255.255.255.0	192.168.1.1

■ TOPOLOGY SETUP

Place: 1 router (1941), Switch0 (2960). Cable: PC0→Fa0/1 Switch0, PC1→Fa0/3 Switch0, Switch0 Fa0/2→R1 Gig0/0.

NOTE: Single router — no OSPF neighbours needed. OSPF MD5 is configured but the main requirement is local AAA.

■ STEPS

Step 1 — Configure IP addressing

```
R1(config)#
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

Step 2 — Configure OSPF MD5

```
R1(config)#
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
area 0 authentication message-digest
exit
interface gig0/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

Step 3 — Create local user FIRST, then enable AAA

CRITICAL: Create username BEFORE typing 'aaa new-model' or you will be locked out.

```
R1(config)#
username Admin1 secret admin1pa55
aaa new-model
aaa authentication login default local
```

Step 4 — Apply AAA to console line

```
R1(config)#
line console 0
login authentication default
exit
```

Step 5 — Configure SSH and apply AAA to VTY lines

```
R1(config)#
ip domain-name ccnasecurity.com
crypto key generate rsa
! Enter 1024 when prompted
aaa authentication login SSH-LOGIN local
line vty 0 4
login authentication SSH-LOGIN
transport input ssh
exit
```

■ VERIFICATION

Console Login: Exit R1 console, press Enter

→ Username: Admin1 Password: admin1pa55 → R1>

SSH from PC0: ssh -l Admin1 192.168.1.1

→ Password prompt → R1>

■ KEY POINTS

- Username 'Admin1' — capital A matters. AAA is case-sensitive.
- 'aaa new-model' takes effect immediately — always create user first.

QUESTION 5

OSPF MD5 + NTP + Syslog + SSH on R3

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
R1	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
R2	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
R3	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A (NTP)	NIC	192.168.1.5	255.255.255.0	192.168.1.1
PC-B (Syslog)	NIC	192.168.1.6	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1

■ TOPOLOGY SETUP

Place devices: 3 routers (1941), Switch0 (2960), PCA, PCB, PCC. Cable: PCA→Switch0→R1 Gig0/0, R1 se0/1/0↔R2 se0/1/0 (DCE on R1), R2 se0/1/1↔R3 se0/1/0 (DCE on R2), R3 Gig0/0→Switch1, PCC→Switch1.

■ STEPS

Step 1 — Configure IPs on all routers

```
R1(config)#
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.1.1.1 255.255.255.252
clock rate 128000
no shutdown
exit
```

```
R2(config)#
interface se0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
interface se0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
```

```
R3(config)#
interface gig0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
exit
```

Step 2 — Configure OSPF on all routers

```
R1(config)#
```

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.3 area 0
exit
```

```
R2(config)#
```

```
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

```
R3(config)#
```

```
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

Step 3 — Add OSPF MD5 on all routers (after FULL adjacency)

```
R1(config)#
```

```
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

```
R2(config)#
```

```
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
interface se0/1/1
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

```
R3(config)#
```

```
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

Step 4 — Enable NTP auth on PC-A (NTP Server)

Click PC-A → Services → NTP → On. Authentication: Enable, Key=1, Password=NTPpa55.

Step 5 — Configure NTP on all routers

```
R1(config)#
```

```
ntp server 192.168.1.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
ntp authentication-key 1 md5 NTPpa55
service timestamps log datetime msec
```

```
R2(config)#
```

```
ntp server 192.168.1.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
ntp authentication-key 1 md5 NTPpa55
service timestamps log datetime msec
```

```
R3(config)#
ntp server 192.168.1.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
ntp authentication-key 1 md5 NTPpa55
service timestamps log datetime msec
```

Step 6 — Configure Syslog on all routers (PC-B is Syslog server)

Click PC-B → Services → SYSLOG → On.

```
R1(config)#
logging host 192.168.1.6
```

```
R2(config)#
logging host 192.168.1.6
```

```
R3(config)#
logging host 192.168.1.6
```

Step 7 — Configure SSH on R3

```
R3(config)#
ip domain-name ccnasecurity.com
username SSHadmin privilege 15 secret ciscosshpa55
crypto key zeroize rsa
! Press Enter if prompted
crypto key generate rsa
! Enter 1024 when prompted for modulus size
ip ssh version 2
ip ssh time-out 90
ip ssh authentication-retries 2
line vty 0 4
login local
transport input ssh
exit
```

■ VERIFICATION

```
R3# show ip ssh → SSH Enabled - version 2.0 timeout 90 retries 2
PC-C> ssh -l SSHadmin 192.168.3.1 → Password: ciscosshpa55 → R3#
PC-C> telnet 192.168.3.1 → Connection refused (Telnet blocked – correct)
```

■ KEY POINTS

- Hostname must be set (not 'Router') before RSA key generation.
- 'transport input ssh' blocks Telnet — a refused Telnet confirms it is working.

QUESTION 6

Extended ACL — SSH from PCB subnet only, ICMP all

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
RTA	gig0/0	10.101.117.49	255.255.255.248	N/A
RTA	gig0/1	10.101.117.33	255.255.255.240	N/A
RTA	gig0/2	10.101.117.1	255.255.255.224	N/A
PCA	NIC	10.101.117.51	255.255.255.248	10.101.117.49
PCB	NIC	10.101.117.35	255.255.255.240	10.101.117.33
SWA	VLAN1	10.101.117.50	255.255.255.248	10.101.117.49
SWB	VLAN1	10.101.117.34	255.255.255.240	10.101.117.33
SWC	VLAN1	10.101.117.2	255.255.255.224	10.101.117.1

■ TOPOLOGY SETUP

Place: 1 router RTA (1941 or 2911), 3 switches SWA/SWB/SWC (2960). Cable: PCA→SWA→RTA Gig0/0, PCB→SWB→RTA Gig0/1, SWC→RTA Gig0/2.

■ STEPS

Step 1 — Configure IPs on RTA

```
RTA(config)#
interface gig0/0
ip address 10.101.117.49 255.255.255.248
no shutdown
exit
interface gig0/1
ip address 10.101.117.33 255.255.255.240
no shutdown
exit
interface gig0/2
ip address 10.101.117.1 255.255.255.224
no shutdown
exit
```

Step 2 — Configure SSH on SWC so it can be accessed remotely

```
SWC(config)#
ip domain-name ccna.com
username Admin secret Adminpa55
crypto key generate rsa
! Enter 1024 when prompted
ip ssh version 2
line vty 0 4
login local
transport input ssh
exit
```

Step 3 — Configure ACL 199 on RTA Gig0/2 outbound

Wildcard /28=0.0.0.15 (PCB subnet), /27=0.0.0.31 (SWC network). Applied outbound toward SWC.

```
RTA(config)#
access-list 199 permit tcp 10.101.117.32 0.0.0.15 10.101.117.0 0.0.0.31 eq 22
access-list 199 permit icmp any any
interface gig0/2
ip access-group 199 out
exit
```

■ VERIFICATION

```
PCB> ssh -l Admin 10.101.117.2 → Login success – PCB subnet permitted
```

```
PCA> ssh -l Admin 10.101.117.2 → Timeout – PCA subnet not permitted
```

```
PCA ping SWC → Success – ICMP permitted for all
```

```
RTA# show access-lists 199 → Match counts > 0 on permit lines
```

■ KEY POINTS

- eq 22 = SSH port. eq www = 80. eq ftp = 21.
- ACL on outbound Gig0/2 affects all traffic heading toward SWC subnet.

QUESTION 7

Layer 2 Security — STP Root, PortFast, BPDU Guard, Port Security

■ ADDRESSING TABLE

Device	Interface	IP / Notes	Subnet Mask	Default GW
Central	Gig0/1 uplink	Root bridge – priority	24576	–
SW-1	Fa0/23-24	Root Guard applied here	–	–
SW-2	Fa0/23-24	Root Guard applied here	–	–
SW-A	Fa0/1-4	PortFast + BPDU Guard +	Port Security	–
SW-B	Fa0/1-4	PortFast + BPDU Guard +	Port Security	–

■ TOPOLOGY SETUP

Place 5 switches: Central (3560 or 2960), SW-1, SW-2 (2960), SW-A, SW-B (2960). Central connects to SW-1 and SW-2 via Gig ports. SW-1 and SW-2 connect down to SW-A and SW-B. Hosts C1-C4 connect to SW-A, D1-D4 to SW-B.

■ STEPS

Step 1 — Assign root and secondary bridge

```
Central(config)#  
spanning-tree vlan 1 root primary
```

```
SW-1(config)#  
spanning-tree vlan 1 root secondary
```

Step 2 — PortFast and BPDU Guard on SW-A access ports

```
SW-A(config)#  
interface range fa0/1 - 4  
spanning-tree portfast  
spanning-tree bpduguard enable  
exit
```

Step 2b — PortFast and BPDU Guard on SW-B access ports

```
SW-B(config)#  
interface range fa0/1 - 4  
spanning-tree portfast  
spanning-tree bpduguard enable  
exit
```

Step 3 — Root Guard on SW-1 uplinks

```
SW-1(config)#  
interface range fa0/23 - 24  
spanning-tree guard root  
exit
```

Step 3b — Root Guard on SW-2 uplinks

```
SW-2(config)#  
interface range fa0/23 - 24  
spanning-tree guard root  
exit
```

Step 4 — Port Security on SW-A (host ports + shut unused)

```
SW-A(config)#  
interface range fa0/1 - 22  
switchport mode access  
switchport port-security  
switchport port-security maximum 2
```



```
switchport port-security violation shutdown
switchport port-security mac-address sticky
exit
interface range fa0/5 - 22
shutdown
exit
```

Step 4b — Port Security on SW-B (host ports + shut unused)

```
SW-B(config)#
interface range fa0/1 - 22
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security mac-address sticky
exit
interface range fa0/5 - 22
shutdown
exit
```

■ VERIFICATION

```
Central# show spanning-tree → 'This bridge is the root' Priority 24577
```

```
SW-A# show port-security interface fa0/1 → Enabled Max MAC: 2 Mode: Shutdown
```

```
Ping C1→C2, then show port-security interface fa0/1 → Sticky MAC count = 1
```

■ KEY POINTS

- 'root primary' sets priority 24576. 'secondary' = 28672. Lowest wins.
- PortFast on ACCESS ports only — never trunk or uplink ports.
- 'switchport mode access' must come before 'switchport port-security'.

QUESTION 8

ZPF Firewall on R1

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
R1	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
R2	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
R3	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A (inside)	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-C (outside)	NIC	192.168.3.3	255.255.255.0	192.168.3.1

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch S1 (PC-A side), Switch S2 (PC-C side). Cable serials R1↔R2↔R3 (DCE on R1 and R2). Verify PC-A and PC-C can ping each other before adding ZPF.

■ STEPS

Step 1 — Configure IPs and static routes on all routers

```
R1(config)#
```

```
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.1.1.1 255.255.255.252
clock rate 128000
no shutdown
exit
ip route 10.2.2.0 255.255.255.252 10.1.1.2
ip route 192.168.3.0 255.255.255.0 10.1.1.2
```

```
R2(config)#
```

```
interface se0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
interface se0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
ip route 192.168.1.0 255.255.255.0 10.1.1.1
ip route 192.168.3.0 255.255.255.0 10.2.2.1
```

```
R3(config)#
```

```
interface gig0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
exit
ip route 192.168.1.0 255.255.255.0 10.2.2.2
```

```
ip route 10.1.1.0 255.255.255.252 10.2.2.2
```

Step 2 — Enable security package on R1 and reload

```
R1(config)#  
license boot module c1900 technology-package securityk9  
! Type yes when prompted  
R1#  
copy running-config startup-config  
reload  
! After reload, log back in and enter configure terminal
```

Step 3 — Create security zones, ACL, and class-map

```
R1(config)#  
zone security IN-ZONE  
exit  
zone security OUT-ZONE  
exit  
access-list 101 permit ip 192.168.1.0 0.0.0.255 any  
class-map type inspect match-all IN-NET-CLASS-MAP  
match access-group 101  
exit
```

Step 4 — Policy-map and zone-pair

```
R1(config)#  
policy-map type inspect IN-2-OUT-PMAP  
class type inspect IN-NET-CLASS-MAP  
inspect  
exit  
exit  
zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-ZONE  
service-policy type inspect IN-2-OUT-PMAP  
exit
```

Step 5 — Assign interfaces to zones

```
R1(config)#  
interface gig0/0  
zone-member security IN-ZONE  
exit  
interface se0/1/0  
zone-member security OUT-ZONE  
exit
```

■ VERIFICATION

```
R1# show zone security → IN-ZONE has Gig0/0 OUT-ZONE has Se0/1/0  
PC-A ping PC-C → Success (inside→outside, inspect allows return traffic)  
PC-C ping PC-A → Timeout (outside→inside blocked – no reverse policy)  
R1# show policy-map type inspect zone-pair sessions → Session entries visible
```

■ KEY POINTS

- ZPF order: Zones → ACL → ClassMap → PolicyMap → ZonePair → Assign interfaces.
- 'inspect' = stateful. 'pass' = one-way stateless. 'drop' = block.
- Security package must be installed before zone security commands exist.

QUESTION 9

IPv6 ACLs — Block HTTP and HTTPS on R1

■ ADDRESSING TABLE

Device	Interface	IPv6 Address/Prefix	Default GW
PC1	NIC	2001:DB8:1:10::10/64	FE80::1
PC2	NIC	2001:DB8:1:11::11/64	FE80::1
R1	gig0/0	2001:DB8:1:10::1/64	—
R1	gig0/1	2001:DB8:1:11::1/64	—
R1	se0/1/0	2001:DB8:1:1::1/64	—
R2	se0/1/0	2001:DB8:1:1::2/64	—
R2	se0/1/1	2001:DB8:1:2::2/64	—
R3	gig0/0	2001:DB8:1:30::1/64	—
R3	se0/1/0	2001:DB8:1:2::1/64	—
Server	NIC	2001:DB8:1:30::30/64	FE80::3

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch1 (PC1, PC2 side on R1), Switch2 (Server side on R3). Use serial cables for R1-R2 and R2-R3.

■ STEPS

Step 1 — Enable IPv6 routing and configure all interfaces

```
R1(config)#
```

```
ipv6 unicast-routing
interface gig0/0
ipv6 address 2001:DB8:1:10::1/64
ipv6 address FE80::1 link-local
no shutdown
exit
interface gig0/1
ipv6 address 2001:DB8:1:11::1/64
ipv6 address FE80::1 link-local
no shutdown
exit
interface se0/1/0
ipv6 address 2001:DB8:1:1::1/64
clock rate 128000
no shutdown
exit
```

```
R2(config)#
```

```
ipv6 unicast-routing
interface se0/1/0
ipv6 address 2001:DB8:1:1::2/64
no shutdown
exit
interface se0/1/1
ipv6 address 2001:DB8:1:2::2/64
clock rate 128000
no shutdown
exit
```

```
R3(config)#
```

```
ipv6 unicast-routing
interface gig0/0
ipv6 address 2001:DB8:1:30::1/64
```

```
ipv6 address FE80::3 link-local
no shutdown
exit
interface se0/1/0
ipv6 address 2001:DB8:1:2::1/64
no shutdown
exit
```

Step 2 — Configure IPv6 static routes on all routers

```
R1(config)#
ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:1::2
ipv6 route 2001:DB8:1:2::/64 2001:DB8:1:1::2
```

```
R2(config)#
ipv6 route 2001:DB8:1:10::/64 2001:DB8:1:1::1
ipv6 route 2001:DB8:1:11::/64 2001:DB8:1:1::1
ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:2::1
```

```
R3(config)#
ipv6 route 2001:DB8:1:10::/64 2001:DB8:1:2::2
ipv6 route 2001:DB8:1:11::/64 2001:DB8:1:2::2
```

Step 3 — Block HTTP (port 80) from PC1 subnet — applied on Gig0/0 inbound

```
R1(config)#
ipv6 access-list BLOCK_HTTP
deny tcp any host 2001:DB8:1:30::30 eq www
permit ipv6 any any
exit
interface gig0/0
ipv6 traffic-filter BLOCK_HTTP in
exit
```

Step 4 — Block HTTPS (port 443) from PC2 subnet — applied on Gig0/1 inbound

```
R1(config)#
ipv6 access-list BLOCK_HTTPS
deny tcp any host 2001:DB8:1:30::30 eq 443
permit ipv6 any any
exit
interface gig0/1
ipv6 traffic-filter BLOCK_HTTPS in
exit
```

■ VERIFICATION

```
PC1 browser http://[2001:DB8:1:30::30] → Timeout (HTTP blocked)
PC1 ping Server → Success (ICMP not blocked)
PC2 browser https://[2001:DB8:1:30::30] → Timeout (HTTPS blocked)
R1# show ipv6 access-list → deny lines show match counts > 0
```

■ KEY POINTS

- IPv6 ACLs: always named, always 'ipv6 traffic-filter', always 'ipv6 access-list'.
- eq www = 80. eq 443 = HTTPS. Both are TCP.

QUESTION 10

IP ACLs to Mitigate Attacks

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
R1	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
R2	se0/1/1	10.2.2.2	255.255.255.252	N/A
R2	Lo0	192.168.2.1	255.255.255.0	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
R3	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A (Server)	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-C (Mgmt)	NIC	192.168.3.3	255.255.255.0	192.168.3.1

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch (PC-A side on R1), Switch (PC-C side on R3). Cable serials R1↔R2↔R3 (DCE on R1 and R2). Configure IPs, OSPF routing, and loopback on R2 first, then verify baseline connectivity before applying ACLs.

■ STEPS

Step 1 — Configure IPs and OSPF routing on all routers

```
R1(config)#
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.1.1.1 255.255.255.252
clock rate 128000
no shutdown
exit
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.3 area 0
exit
```

```
R2(config)#
interface se0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
interface se0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
interface loopback 0
ip address 192.168.2.1 255.255.255.0
exit
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
network 192.168.2.0 0.0.0.255 area 0
exit
```

```
R3(config)#
interface gig0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
exit
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

Step 2 — ACL 10 — Restrict VTY access to PC-C only (apply on all routers)

```
R1(config)#
access-list 10 permit host 192.168.3.3
line vty 0 4
access-class 10 in
exit
```

```
R2(config)#
access-list 10 permit host 192.168.3.3
line vty 0 4
access-class 10 in
exit
```

```
R3(config)#
access-list 10 permit host 192.168.3.3
line vty 0 4
access-class 10 in
exit
```

Step 3 — ACL 120 on R1 Se0/1/0 inbound — control external traffic to PC-A

```
R1(config)#
access-list 120 permit udp any host 192.168.1.3 eq domain
access-list 120 permit tcp any host 192.168.1.3 eq smtp
access-list 120 permit tcp any host 192.168.1.3 eq ftp
access-list 120 deny tcp any host 192.168.1.3 eq 443
access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 eq 22
access-list 120 permit icmp any any echo-reply
access-list 120 permit icmp any any unreachable
access-list 120 deny icmp any any
access-list 120 permit ip any any
interface se0/1/0
ip access-group 120 in
exit
```

Step 4 — R3 anti-spoofing ACL 110 on Gig0/0 inbound

```
R3(config)#
access-list 110 permit ip 192.168.3.0 0.0.0.255 any
interface gig0/0
ip access-group 110 in
exit
```

Step 5 — R3 RFC 1918 blocking ACL 100 on Se0/1/0 inbound

```
R3(config)#
access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22 host 192.168.3.3
access-list 100 deny ip 10.0.0.0 0.255.255.255 any
access-list 100 deny ip 172.16.0.0 0.15.255.255 any
access-list 100 deny ip 192.168.0.0 0.0.255.255 any
```

```
access-list 100 deny ip 127.0.0.0 0.255.255.255 any
access-list 100 deny ip 224.0.0.0 15.255.255.255 any
access-list 100 permit ip any any
interface se0/1/0
ip access-group 100 in
exit
```

■ VERIFICATION

```
PC-C telnet any router → Login prompt (permitted)
PC-A telnet any router → Connection refused (blocked by ACL 10)
PC-C ping PC-A → Success (echo-reply permitted by ACL 120)
PC-C HTTPS to PC-A → Fail (port 443 denied)
PC-C FTP/DNS/SMTP to PC-A → Success
```

■ KEY POINTS

- 'access-class' on VTY. 'ip access-group' on interfaces. Never swap these.
- SSH permit **MUST** appear before deny 10.0.0.0/8 in ACL 100 — order matters.
- RFC 1918: 10/8, 172.16/12, 192.168/16 — memorise these.

Security in Computing

Practice Questions — Complete Solutions

From-Scratch Topology Setup · Full CLI · Verification
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QUESTION 11

IOS IPS — Enable, Configure, Modify Signature

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
R1	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
R2	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
R3	se0/1/0	10.2.2.1	255.255.255.252	N/A
Syslog	NIC	192.168.1.50	255.255.255.0	192.168.1.1
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch S1 (Syslog + PC-A side on R1), Switch S2 (PC-C side on R3). Cable serials R1↔R2↔R3 (DCE on R1 and R2). Verify baseline pings in both directions before enabling IPS.

■ STEPS

Step 1 — Configure IPs and OSPF routing on all routers

```
R1(config)#
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.1.1.1 255.255.255.252
clock rate 128000
no shutdown
exit
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.3 area 0
exit
```

```
R2(config)#
interface se0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
interface se0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

```
R3(config)#
interface gig0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
```

```
exit
interface se0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
exit
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

Step 2 — Enable security package on R1 and reload

```
R1(config)#
license boot module c1900 technology-package securityk9
! Type yes when prompted

R1#
copy running-config startup-config
reload
! After reload log back in and enter configure terminal
```

Step 3 — Set clock, create IPS directory, configure IPS rule

```
R1#
clock set 10:20:00 10 january 2014
mkdir ipsdir
! Press Enter to confirm directory creation

R1(config)#
ip ips config location flash:ipsdir
ip ips name iosips
ip ips notify log
service timestamps log datetime msec
logging host 192.168.1.50
```

Step 4 — Configure signature categories (retire all, then enable ios_ips basic)

```
R1(config)#
ip ips signature-category
category all
retired true
exit
category ios_ips basic
retired false
exit
exit
! Press Enter to confirm when prompted
```

Step 5 — Apply IPS rule to Gig0/0 outbound

```
R1(config)#
interface gig0/0
ip ips iosips out
exit
```

Step 6 — Modify signature 2004 — alert and drop ICMP echo requests

```
R1(config)#
ip ips signature-definition
signature 2004 0
status
retired false
enabled true
exit
engine
event-action produce-alert
event-action deny-packet-inline
```

```
exit
exit
exit
! Press Enter to confirm when prompted
```

■ VERIFICATION

```
R1# show ip ips interfaces
→ GigabitEthernet0/0 Outbound: iosips
PC-C ping PC-A → Timeout (ICMP echo request type 8 dropped by sig 2004)
PC-A ping PC-C → Success (only REQUEST is matched, not REPLY)
Syslog server → SYSLOG tab → %IPS-4-SIGNATURE: Sig:2004 entries visible
```

■ KEY POINTS

- Sig 2004 = echo REQUEST. Sig 2000 = echo REPLY. Do not modify the wrong one.
- 'out' on Gig0/0 = inspect traffic going toward the LAN from the WAN direction.
- Retire ALL first, then un-retire ios_ips basic — thousands of active sigs overload the router.

QUESTION 12

ZPF Firewall on R3

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
R1	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
R2	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
R3	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A (outside)	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-C (inside)	NIC	192.168.3.3	255.255.255.0	192.168.3.1

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch S1 (PC-A side on R1), Switch S2 (PC-C side on R3). Cable serials R1↔R2↔R3. Gig0/0 on R3 (facing PC-C LAN) → IN-ZONE. Se0/1/0 on R3 (facing WAN) → OUT-ZONE.

■ STEPS

Step 1 — Configure IPs and static routes on all routers

```
R1(config)#
```

```
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.1.1.1 255.255.255.252
clock rate 128000
no shutdown
exit
ip route 10.2.2.0 255.255.255.252 10.1.1.2
ip route 192.168.3.0 255.255.255.0 10.1.1.2
```

```
R2(config)#
```

```
interface se0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
interface se0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
ip route 192.168.1.0 255.255.255.0 10.1.1.1
ip route 192.168.3.0 255.255.255.0 10.2.2.1
```

```
R3(config)#
```

```
interface gig0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
exit
ip route 192.168.1.0 255.255.255.0 10.2.2.2
```

```
ip route 10.1.1.0 255.255.255.252 10.2.2.2
```

Step 2 — Enable security package on R3 and reload

```
R3(config)#  
license boot module c1900 technology-package securityk9  
! Type yes when prompted
```

```
R3#  
copy running-config startup-config  
reload  
! After reload log back in and enter configure terminal
```

Step 3 — Create security zones, ACL, and class-map on R3

The inside LAN on R3's side is 192.168.3.0/24.

```
R3(config)#  
zone security IN-ZONE  
exit  
zone security OUT-ZONE  
exit  
access-list 101 permit ip 192.168.3.0 0.0.0.255 any  
class-map type inspect match-all IN-NET-CLASS-MAP  
match access-group 101  
exit
```

Step 4 — Policy-map and zone-pair on R3

```
R3(config)#  
policy-map type inspect IN-2-OUT-PMAP  
class type inspect IN-NET-CLASS-MAP  
inspect  
exit  
exit  
zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-ZONE  
service-policy type inspect IN-2-OUT-PMAP  
exit
```

Step 5 — Assign interfaces to zones on R3

```
R3(config)#  
interface gig0/0  
zone-member security IN-ZONE  
exit  
interface se0/1/0  
zone-member security OUT-ZONE  
exit
```

■ VERIFICATION

```
R3# show zone security → IN-ZONE has Gig0/0 OUT-ZONE has Se0/1/0  
PC-C ping PC-A (inside→outside) → Success  
PC-A ping PC-C (outside→inside) → Timeout – no reverse policy
```

■ KEY POINTS

- ZPF order: Zones → ACL → ClassMap → PolicyMap → ZonePair → Assign interfaces.
- 'inspect' = stateful (allows return traffic). 'pass' = one-way stateless. 'drop' = block.
- Security package must be installed before zone security commands exist.

QUESTION 13

OSPF MD5 + NTP + Syslog (NTP/Syslog on R3 LAN)

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
R1	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
R2	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
R3	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1
PC-B (NTP Server)	NIC	192.168.3.5	255.255.255.0	192.168.3.1
PC-C (Syslog)	NIC	192.168.3.6	255.255.255.0	192.168.3.1

Note: NTP and Syslog servers are on R3's LAN (192.168.3.x). NTP server = PCB (192.168.3.5), Syslog = PCC (192.168.3.6). All routers must have OSPF routes to 192.168.3.0/24 to reach both servers.

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch0 (PC-A side on R1), Switch1 (PCB + PCC side on R3). Cable: PC-A→Switch0→R1 Gig0/0, R1 se0/1/0↔R2 se0/1/0 (DCE on R1), R2 se0/1/1↔R3 se0/1/0 (DCE on R2), PCB→Switch1, PCC→Switch1, Switch1→R3 Gig0/0.

■ STEPS

Step 1 — Configure IPs on all routers

```
R1(config)#
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.1.1.1 255.255.255.252
clock rate 128000
no shutdown
exit
```

```
R2(config)#
interface se0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
interface se0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
```

```
R3(config)#
interface gig0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
exit
```

Step 2 — Configure OSPF on all routers

```
R1(config)#
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.3 area 0
exit
```

```
R2(config)#
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

```
R3(config)#
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

Step 3 — Add OSPF MD5 on all routers (after FULL adjacency confirmed)

```
R1(config)#
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

```
R2(config)#
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
interface se0/1/1
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

```
R3(config)#
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

Step 4 — Enable NTP auth on PC-B (NTP Server on R3 LAN)

Click PC-B → Services → NTP → On. Authentication: Enable, Key=1, Password=NTPPa55.

Step 5 — Configure NTP on all routers

```
R1(config)#
ntp server 192.168.3.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
ntp authentication-key 1 md5 NTPPa55
service timestamps log datetime msec
```

```
R2(config)#
ntp server 192.168.3.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
```



```
ntp authentication-key 1 md5 NTPpa55
service timestamps log datetime msec
```

```
R3(config)#
ntp server 192.168.3.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
ntp authentication-key 1 md5 NTPpa55
service timestamps log datetime msec
```

Step 6 — Configure Syslog on all routers (PC-C is Syslog server)

Click PC-C → Services → SYSLOG → On.

```
R1(config)#
logging host 192.168.3.6
```

```
R2(config)#
logging host 192.168.3.6
```

```
R3(config)#
logging host 192.168.3.6
```

■ VERIFICATION

```
R1# show ntp status → synchronized, stratum 2, reference 192.168.3.5
R1# show ip ospf interface se0/1/0 → Message digest authentication enabled
Enter/exit config on R1, check PC-C SYSLOG tab → %SYS-5-CONFIG_I entries appear
```

■ KEY POINTS

- NTP/Syslog servers are on R3's LAN — all routers need OSPF route to 192.168.3.0/24.
- Identify WHICH PC is NTP vs Syslog from the topology before starting config.

QUESTION 14

IOS IPS — Syslog Server on R3 LAN

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
R1	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0 (DCE)	10.1.1.2	255.255.255.252	N/A
R2	se0/1/1 (DCE)	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
R3	se0/1/0	10.2.2.1	255.255.255.252	N/A
Syslog	NIC	192.168.3.50	255.255.255.0	192.168.3.1
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1

Note: Syslog server is on R3's LAN (192.168.3.50). R1 must have a route to 192.168.3.0/24 for logs to arrive at the server.

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch S1 (PC-A side on R1), Switch S2 (Syslog + PC-C side on R3). Cable serials R1↔R2↔R3 (DCE on R2 for both serial links). Verify baseline connectivity before enabling IPS.

■ STEPS

Step 1 — Configure IPs and OSPF routing on all routers

```
R1(config)#
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.1.1.1 255.255.255.252
no shutdown
exit
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.3 area 0
exit
```

```
R2(config)#
interface se0/1/0
ip address 10.1.1.2 255.255.255.252
clock rate 128000
no shutdown
exit
interface se0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

```
R3(config)#
interface gig0/0
ip address 192.168.3.1 255.255.255.0
```

```
no shutdown
exit
interface se0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
exit
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

Step 2 — Enable security package on R1 and reload

```
R1(config)#
license boot module c1900 technology-package securityk9
! Type yes when prompted

R1#
copy running-config startup-config
reload
! After reload log back in and enter configure terminal
```

Step 3 — Set clock, create IPS directory, configure IPS rule

```
R1#
clock set 10:20:00 10 january 2014
mkdir ipsdir
! Press Enter to confirm

R1(config)#
ip ips config location flash:ipsdir
ip ips name iosips
ip ips notify log
service timestamps log datetime msec
logging host 192.168.3.50
! Syslog server is on R3 LAN — IP is 192.168.3.50
```

Step 4 — Configure signature categories

```
R1(config)#
ip ips signature-category
category all
retired true
exit
category ios_ips basic
retired false
exit
exit
! Press Enter to confirm
```

Step 5 — Apply IPS rule to Gig0/0 outbound

```
R1(config)#
interface gig0/0
ip ips iosips out
exit
```

Step 6 — Modify signature 2004 — alert and drop ICMP echo requests

```
R1(config)#
ip ips signature-definition
signature 2004 0
status
retired false
enabled true
exit
engine
```

```
event-action produce-alert
event-action deny-packet-inline
exit
exit
exit
! Press Enter to confirm
```

■ VERIFICATION

```
R1# show ip ips interfaces → GigabitEthernet0/0 Outbound: iosips
```

```
PC-C ping PC-A → Timeout (ICMP dropped by sig 2004)
```

```
Syslog server (192.168.3.50) → SYSLOG tab → %IPS-4-SIGNATURE: Sig:2004 visible
```

■ KEY POINTS

- Syslog is on R3's LAN — R1 must have a route to 192.168.3.0/24 for logs to arrive.
- Sig 2004 = ICMP echo REQUEST. Do not confuse with sig 2000 (echo REPLY).

QUESTION 17

OSPF MD5 + NTP + Syslog + SSH on R1

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
R1	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
R2	se0/1/1	10.2.2.2	255.255.255.252	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
R3	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1
PC-B (NTP Server)	NIC	192.168.3.5	255.255.255.0	192.168.3.1
PC-C (Syslog)	NIC	192.168.3.6	255.255.255.0	192.168.3.1

Note: NTP and Syslog servers are on R3's LAN. NTP = PCB (192.168.3.5), Syslog = PCC (192.168.3.6). SSH is configured on R1.

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch0 (PC-A side on R1), Switch1 (PCB + PCC side on R3). Cable serials R1↔R2↔R3 (DCE on R1 and R2). PC-A connects to Switch0, PCB and PCC connect to Switch1.

■ STEPS

Step 1 — Configure IPs on all routers

```
R1(config)#
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.1.1.1 255.255.255.252
clock rate 128000
no shutdown
exit
```

```
R2(config)#
interface se0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
interface se0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
```

```
R3(config)#
interface gig0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
exit
```

Step 2 — Configure OSPF on all routers

```
R1(config)#
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.3 area 0
exit
```

```
R2(config)#
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

```
R3(config)#
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

Step 3 — Add OSPF MD5 on all routers (after FULL adjacency confirmed)

```
R1(config)#
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

```
R2(config)#
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
interface se0/1/1
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

```
R3(config)#
router ospf 1
area 0 authentication message-digest
exit
interface se0/1/0
ip ospf message-digest-key 1 md5 MD5pa55
exit
```

Step 4 — Enable NTP auth on PC-B (NTP Server on R3 LAN)

Click PC-B → Services → NTP → On. Authentication: Enable, Key=1, Password=NTPpa55.

Step 5 — Configure NTP on all routers

```
R1(config)#
ntp server 192.168.3.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
ntp authentication-key 1 md5 NTPpa55
service timestamps log datetime msec
```

```
R2(config)#
ntp server 192.168.3.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
ntp authentication-key 1 md5 NTPpa55
```

```
service timestamps log datetime msec
```

```
R3(config)#
```

```
ntp server 192.168.3.5
ntp update-calendar
ntp authenticate
ntp trusted-key 1
ntp authentication-key 1 md5 NTPpa55
service timestamps log datetime msec
```

Step 6 — Configure Syslog on all routers (PC-C is Syslog server)

Click PC-C → Services → SYSLOG → On.

```
R1(config)#
```

```
logging host 192.168.3.6
```

```
R2(config)#
```

```
logging host 192.168.3.6
```

```
R3(config)#
```

```
logging host 192.168.3.6
```

Step 7 — Configure SSH on R1

```
R1(config)#
```

```
ip domain-name ccnasecurity.com
username SSHadmin privilege 15 secret ciscosshpa55
crypto key zeroize rsa
! Press Enter if prompted
crypto key generate rsa
! Enter 1024 when prompted for modulus size
ip ssh version 2
ip ssh time-out 90
ip ssh authentication-retries 2
line vty 0 4
login local
transport input ssh
exit
```

■ VERIFICATION

```
R1# show ip ssh → SSH Enabled - version 2.0 timeout 90 retries 2
```

```
PC-A> ssh -l SSHadmin 192.168.1.1 → Password: ciscosshpa55 → R1#
```

```
PC-A> telnet 192.168.1.1 → Connection refused (correct – Telnet blocked)
```

```
R1# show ntp status → synchronized, stratum 2, reference 192.168.3.5
```

■ KEY POINTS

- Hostname must be set (not 'Router') before RSA key generation.
- 'transport input ssh' blocks Telnet. A refused Telnet connection confirms it is working.
- NTP server is on R3 LAN — verify OSPF route to 192.168.3.0/24 exists on R1 before testing NTP.

QUESTION 18

IP ACLs to Mitigate Attacks

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/0	192.168.1.1	255.255.255.0	N/A
R1	se0/1/0	10.1.1.1	255.255.255.252	N/A
R2	se0/1/0	10.1.1.2	255.255.255.252	N/A
R2	se0/1/1	10.2.2.2	255.255.255.252	N/A
R2	Lo0	192.168.2.1	255.255.255.0	N/A
R3	gig0/0	192.168.3.1	255.255.255.0	N/A
R3	se0/1/0	10.2.2.1	255.255.255.252	N/A
PC-A (Server)	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-C (Mgmt)	NIC	192.168.3.3	255.255.255.0	192.168.3.1

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch (PC-A side on R1), Switch (PC-C side on R3). R2 Lo0 simulates ISP. Cable serials R1↔R2↔R3. Configure all IPs including loopback, OSPF, and verify connectivity before applying ACLs.

■ STEPS

Step 1 — Configure IPs, loopback, and OSPF on all routers

```
R1(config)#
```

```
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.1.1.1 255.255.255.252
clock rate 128000
no shutdown
exit
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.3 area 0
exit
```

```
R2(config)#
```

```
interface se0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
interface se0/1/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
interface loopback 0
ip address 192.168.2.1 255.255.255.0
exit
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
network 192.168.2.0 0.0.0.255 area 0
exit
```



```
R3(config)#
interface gig0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
interface se0/1/0
ip address 10.2.2.1 255.255.255.252
no shutdown
exit
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

Step 2 — ACL 10 — Restrict VTY access to PC-C only (apply on all routers)

```
R1(config)#
access-list 10 permit host 192.168.3.3
line vty 0 4
access-class 10 in
exit
```

```
R2(config)#
access-list 10 permit host 192.168.3.3
line vty 0 4
access-class 10 in
exit
```

```
R3(config)#
access-list 10 permit host 192.168.3.3
line vty 0 4
access-class 10 in
exit
```

Step 3 — ACL 120 on R1 Se0/1/0 inbound — control external traffic to PC-A

```
R1(config)#
access-list 120 permit udp any host 192.168.1.3 eq domain
access-list 120 permit tcp any host 192.168.1.3 eq smtp
access-list 120 permit tcp any host 192.168.1.3 eq ftp
access-list 120 deny tcp any host 192.168.1.3 eq 443
access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 eq 22
access-list 120 permit icmp any any echo-reply
access-list 120 permit icmp any any unreachable
access-list 120 deny icmp any any
access-list 120 permit ip any any
interface se0/1/0
ip access-group 120 in
exit
```

Step 4 — R3 anti-spoofing ACL 110 on Gig0/0 inbound

```
R3(config)#
access-list 110 permit ip 192.168.3.0 0.0.0.255 any
interface gig0/0
ip access-group 110 in
exit
```

Step 5 — R3 RFC 1918 blocking ACL 100 on Se0/1/0 inbound

```
R3(config)#
access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22 host 192.168.3.3
access-list 100 deny ip 10.0.0.0 0.255.255.255 any
access-list 100 deny ip 172.16.0.0 0.15.255.255 any
access-list 100 deny ip 192.168.0.0 0.0.255.255 any
```

```
access-list 100 deny ip 127.0.0.0 0.255.255.255 any
access-list 100 deny ip 224.0.0.0 15.255.255.255 any
access-list 100 permit ip any any
interface se0/1/0
ip access-group 100 in
exit
```

■ VERIFICATION

```
PC-C telnet any router → Login prompt (permitted)
PC-A telnet any router → Connection refused (blocked by ACL 10)
PC-C ping PC-A → Success (echo-reply permitted by ACL 120)
PC-C HTTPS to PC-A → Fail (port 443 denied)
PC-C FTP/DNS/SMTP to PC-A → Success
```

■ KEY POINTS

- 'access-class' on VTY. 'ip access-group' on interfaces. Never swap these.
- SSH permit **MUST** appear before deny 10.0.0.0/8 in ACL 100 — order matters.
- RFC 1918: 10/8, 172.16/12, 192.168/16 — memorise these.

QUESTION 19

Extended ACL — SSH from PCB subnet only, ICMP all

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
RTA	gig0/0	10.101.117.49	255.255.255.248	N/A
RTA	gig0/1	10.101.117.33	255.255.255.240	N/A
RTA	gig0/2	10.101.117.1	255.255.255.224	N/A
PCA	NIC	10.101.117.51	255.255.255.248	10.101.117.49
PCB	NIC	10.101.117.35	255.255.255.240	10.101.117.33
SWA	VLAN1	10.101.117.50	255.255.255.248	10.101.117.49
SWB	VLAN1	10.101.117.34	255.255.255.240	10.101.117.33
SWC	VLAN1	10.101.117.2	255.255.255.224	10.101.117.1

■ TOPOLOGY SETUP

Place: 1 router RTA (1941 or 2911), 3 switches SWA/SWB/SWC (2960). Cable: PCA→SWA→RTA Gig0/0, PCB→SWB→RTA Gig0/1, SWC→RTA Gig0/2.

■ STEPS

Step 1 — Configure IPs on RTA

```
RTA(config)#
interface gig0/0
ip address 10.101.117.49 255.255.255.248
no shutdown
exit
interface gig0/1
ip address 10.101.117.33 255.255.255.240
no shutdown
exit
interface gig0/2
ip address 10.101.117.1 255.255.255.224
no shutdown
exit
```

Step 2 — Configure SSH on SWC

```
SWC(config)#
ip domain-name ccna.com
username Admin secret Adminpa55
crypto key generate rsa
! Enter 1024 when prompted
ip ssh version 2
line vty 0 4
login local
transport input ssh
exit
```

Step 3 — Configure ACL 199 on RTA Gig0/2 outbound

Wildcard /28=0.0.0.15 (PCB subnet), /27=0.0.0.31 (SWC network). Applied outbound toward SWC.

```
RTA(config)#
access-list 199 permit tcp 10.101.117.32 0.0.0.15 10.101.117.0 0.0.0.31 eq 22
access-list 199 permit icmp any any
interface gig0/2
ip access-group 199 out
exit
```

■ VERIFICATION

```
PCB> ssh -l Admin 10.101.117.2 → Login success – PCB subnet permitted
```

```
PCA> ssh -l Admin 10.101.117.2 → Timeout – PCA subnet not permitted
```

```
PCA ping SWC → Success – ICMP permitted for all
```

```
RTA# show access-lists 199 → Match counts > 0 on permit lines
```

■ KEY POINTS

- eq 22 = SSH port. Wildcard /28 = 0.0.0.15 (PCB subnet). Wildcard /27 = 0.0.0.31 (SWC network).
- ACL on outbound Gig0/2 affects all traffic heading toward SWC subnet.

QUESTION 20

IOS IPS — Alternate Interface Names

■ ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default GW
R1	gig0/1	192.168.1.1	255.255.255.0	N/A
R1	se0/0/0	10.1.1.1	255.255.255.252	N/A
R2	se0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A
R2	se0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A
R3	gig0/1	192.168.3.1	255.255.255.0	N/A
R3	se0/0/0	10.2.2.1	255.255.255.252	N/A
Syslog	NIC	192.168.1.50	255.255.255.0	192.168.1.1
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1

Note: Interface names differ from Q11 — Gig0/1 instead of Gig0/0, se0/0/0 instead of se0/1/0. All IPS logic is identical.

■ STEPS

Step 1 — Configure IPs and OSPF routing on all routers

```
R1(config)#
```

```
interface gig0/1
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface se0/0/0
ip address 10.1.1.1 255.255.255.252
no shutdown
exit
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.1.1.0 0.0.0.3 area 0
exit
```

```
R2(config)#
```

```
interface se0/0/0
ip address 10.1.1.2 255.255.255.252
clock rate 128000
no shutdown
exit
interface se0/0/1
ip address 10.2.2.2 255.255.255.252
clock rate 128000
no shutdown
exit
router ospf 1
network 10.1.1.0 0.0.0.3 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

```
R3(config)#
```

```
interface gig0/1
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
interface se0/0/0
ip address 10.2.2.1 255.255.255.252
```

```
no shutdown
exit
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.3 area 0
exit
```

Step 2 — Enable security package on R1 and reload

```
R1(config)#
license boot module c1900 technology-package securityk9
! Type yes when prompted

R1#
copy running-config startup-config
reload
! After reload log back in and enter configure terminal
```

Step 3 — Set clock, create IPS directory, configure IPS rule

```
R1#
clock set 10:20:00 10 january 2014
mkdir ipsdir
! Press Enter to confirm

R1(config)#
ip ips config location flash:ipsdir
ip ips name iosips
ip ips notify log
service timestamps log datetime msec
logging host 192.168.1.50
```

Step 4 — Configure signature categories

```
R1(config)#
ip ips signature-category
category all
retired true
exit
category ios_ips basic
retired false
exit
exit
! Press Enter to confirm
```

Step 5 — Apply IPS rule to Gig0/1 outbound (note: gig0/1 not gig0/0)

```
R1(config)#
interface gig0/1
ip ips iosips out
exit
```

Step 6 — Modify signature 2004

```
R1(config)#
ip ips signature-definition
signature 2004 0
status
retired false
enabled true
exit
engine
event-action produce-alert
event-action deny-packet-inline
exit
exit
exit
```

! Press Enter to confirm

■ VERIFICATION

R1# show ip ips interfaces → GigabitEthernet0/1 Outbound: iosips

PC-C ping PC-A → Timeout (ICMP dropped by sig 2004)

PC-A ping PC-C → Success (reply not blocked)

Syslog (192.168.1.50) → SYSLOG tab → %IPS-4-SIGNATURE: Sig:2004 visible

■ KEY POINTS

- Only interface names change vs Q11. 'ip ips iosips out' goes on Gig0/1 (not Gig0/0).
- Serial interfaces are se0/0/0 and se0/0/1 in this question — adjust accordingly.

QUESTION 21

IPv6 ACLs — Block HTTP on R3, Block HTTPS on R3

■ ADDRESSING TABLE

Device	Interface	IPv6 Address/Prefix	Default GW
Server	NIC	2001:DB8:1:10::30/64	FE80::1
R1	gig0/0	2001:DB8:1:10::1/64	—
R1	se0/1/0	2001:DB8:1:1::1/64	—
R2	se0/1/0	2001:DB8:1:1::2/64	—
R2	se0/1/1	2001:DB8:1:2::2/64	—
R3	gig0/0	2001:DB8:1:30::1/64	—
R3	gig0/1	2001:DB8:1:31::1/64	—
R3	se0/1/0	2001:DB8:1:2::1/64	—
PCA	NIC	2001:DB8:1:30::10/64	FE80::3
PCB	NIC	2001:DB8:1:31::11/64	FE80::3

■ TOPOLOGY SETUP

Place: 3 routers (1941), Switch S1 (Server side on R1 Gig0/0), Switch S2 (PCA+PCB side on R3). Enable IPv6 unicast-routing on all routers. Set all addresses. Use static routes for routing.

■ STEPS

Step 1 — Enable IPv6 routing and configure all interfaces

```
R1(config)#
```

```
ipv6 unicast-routing
interface gig0/0
ipv6 address 2001:DB8:1:10::1/64
ipv6 address FE80::1 link-local
no shutdown
exit
interface se0/1/0
ipv6 address 2001:DB8:1:1::1/64
clock rate 128000
no shutdown
exit
```

```
R2(config)#
```

```
ipv6 unicast-routing
interface se0/1/0
ipv6 address 2001:DB8:1:1::2/64
no shutdown
exit
interface se0/1/1
ipv6 address 2001:DB8:1:2::2/64
clock rate 128000
no shutdown
exit
```

```
R3(config)#
```

```
ipv6 unicast-routing
interface gig0/0
ipv6 address 2001:DB8:1:30::1/64
ipv6 address FE80::3 link-local
no shutdown
exit
interface gig0/1
```



```
ipv6 address 2001:DB8:1:31::1/64
ipv6 address FE80::3 link-local
no shutdown
exit
interface se0/1/0
ipv6 address 2001:DB8:1:2::1/64
no shutdown
exit
```

Step 2 — Configure IPv6 static routes on all routers

```
R1(config)#
ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:1::2
ipv6 route 2001:DB8:1:31::/64 2001:DB8:1:1::2
ipv6 route 2001:DB8:1:2::/64 2001:DB8:1:1::2
```

```
R2(config)#
ipv6 route 2001:DB8:1:10::/64 2001:DB8:1:1::1
ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:2::1
ipv6 route 2001:DB8:1:31::/64 2001:DB8:1:2::1
```

```
R3(config)#
ipv6 route 2001:DB8:1:10::/64 2001:DB8:1:2::2
ipv6 route 2001:DB8:1:1::/64 2001:DB8:1:2::2
```

Step 3 — BLOCK_HTTP on R3 — block HTTP (port 80) to Server (inbound on Gig0/0, PCA side)

```
R3(config)#
ipv6 access-list BLOCK_HTTP
deny tcp any host 2001:DB8:1:10::30 eq www
permit ipv6 any any
exit
interface gig0/0
ipv6 traffic-filter BLOCK_HTTP in
exit
```

Step 4 — BLOCK_HTTPS on R3 — block HTTPS (port 443) to Server (inbound on Gig0/1, PCB side)

```
R3(config)#
ipv6 access-list BLOCK_HTTPS
deny tcp any host 2001:DB8:1:10::30 eq 443
permit ipv6 any any
exit
interface gig0/1
ipv6 traffic-filter BLOCK_HTTPS in
exit
```

■ VERIFICATION

```
PCA browser http://[2001:DB8:1:10::30] → Timeout (HTTP blocked)
PCA ping Server → Success (ICMP not blocked)
PCB browser https://[2001:DB8:1:10::30] → Timeout (HTTPS blocked)
PCB HTTP to Server → Success (only HTTPS blocked on that ACL)
R3# show ipv6 access-list → deny lines show match counts > 0
```

■ KEY POINTS

- eq www = HTTP port 80. eq 443 = HTTPS. Both are TCP.
- Separate ACLs on separate interfaces gives finer control per subnet.
- IPv6 ACLs: always named, 'ipv6 traffic-filter' not 'ip access-group'.